

		$\lambda = 300 \text{ m}$
13	B	Amplitude $x_0 = 600 / 4 = 150 \text{ mm} = 0.150 \text{ m}$ Frequency $f = 50 / 47 = 1.064 \text{ Hz}$ Angular frequency $\omega = 2\pi f = 2(\pi)(1.064) = 6.684 \text{ rad s}^{-1}$ Max KE = $\frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} \times (5.0 \times 10^{-3}) \times (6.684)^2 \times (0.150)^2 = 2.51 \times 10^{-3} \text{ J}$
14	B	$v = f \lambda$